Evaluation report - X sentiment price prediction for Bitcoin (BTC/USD)

CrowdWisdomTrading quantitative data science intern task. Question: can daily X (Twitter) sentiment about Bitcoin, combined with price history only, forecast the BTC/USD price 0-5 days ahead?

Data

- Posts: 70,585 Bitcoin tweets, 2022-08-30 to 2023-01-09, from the public Kaggle 'Bitcoin Tweets' corpus (kaushiksuresh147), mirrored on Hugging Face as khaihernlow/bitcoin-tweets. The corpus has author follower counts but no per-tweet like/repost counts, so engagement weighting uses followers (documented in the README). Stored volume was capped at 2,000 posts/day to keep scoring time sane. The corpus was collected in bursts, so it covers 36 collection days inside the window; on other days sentiment is carried forward and post volume is 0 (see Limitations).

- Prices: 133 daily BTC/USD OHLCV bars from the Coinbase Exchange public API for the same window.

- Sentiment: VADER compound per post. Overall mean +0.139, 42% of posts positive, 16% negative.

- Posts and prices are loaded into SQLite (data/cwt.db, tables posts, prices, daily_features).

Method

Daily features are built from sentiment + price only: sentiment level/lags/rolling means/momentum, engagement-weighted sentiment, post volume and its z-score, positive/negative share, lagged returns and rolling volatility. Price features are lagged one day so nothing from the future leaks in. One GradientBoostingRegressor per horizon predicts the return to n+0 .. n+5.

Validation is walk-forward: expanding window, first 60% of days as initial training, then refit every 14 days and score the next 14. The baseline is persistence (predict tomorrow = today).

Forecast results (out-of-sample)

Horizon	Model RMSE	Baseline RMSE	Model MAE	Baseline MAE	Directional acc.
n+0	555	240	421	164	47.9%
n+1	438	240	352	164	45.8%
n+2	608	344	485	229	47.9%
n+3	605	396	495	286	55.3%
n+4	796	414	647	304	55.3%
n+5	1,054	437	822	343	34.8%

RMSE/MAE are in USD on the predicted price. Directional accuracy is the share of days where the predicted return had the right sign (the persistence baseline predicts a flat return, so it has no direction to score).

Backtest (long/flat on the n+1 signal)

Out-of-sample period 2022-11-22 to 2023-01-08 (48 days, in the market 60% of the time, only trading when the predicted return clears +0.2%):

| Metric | Strategy | Buy & hold |

|---|---|---|

| Total return | +5.7% | +6.0% |

| Sharpe ratio | 1.76 | - |

| Sortino ratio | 1.69 | - |

| Max drawdown | -6.6% | - |

| Win rate (active days) | 45% | - |

No transaction costs or slippage are modelled, and the strategy never shorts, so these numbers are optimistic.

Charts

- charts/price_vs_sentiment.png - price vs sentiment over time

- charts/sentiment_distribution.png - per-post sentiment distribution

- charts/predicted_vs_actual.png - walk-forward forecast vs actual (n+1)

- charts/equity_curve.png - backtest equity curve and drawdown

Limitations

- Data period: public tweet corpora with usable timestamps mostly end in early 2023 (free X data collection got much harder after the API changes), so the offline run uses Aug 2022 - Jan 2023. The live Apify mode in src/ingest_apify.py covers the recent 180 days when an API token is available.

- No per-tweet engagement in the offline corpus (no likes or reposts), so engagement features lean on author followers.

- VADER is a general lexicon, not finance-specific; it scores sarcasm and crypto slang imperfectly.

- One asset, one window. Results may not hold in other market regimes; BTC is close to a random walk at daily frequency, which is why the naive baseline is hard to beat on RMSE.

- The 2,000 posts/day cap means volume features describe the sampled stream, not the full firehose.

- Burst collection: the public corpus was gathered a few weeks at a time, so only 36 of the 133 window days have posts. Sentiment is forward-filled across the gaps (volume = 0 there), which weakens the daily signal - another reason the live Apify mode is preferable when a token is available.